DS	**Session**	Hashing	**Question Information**	Level 1 Challenge 94

Question description

The students of college XYZ are getting jealous of the students of college ABC. ABC managed to beat XYZ in all the sports and games events. The main strength of the students of ABC is their unity. The students of XYZ decide to destroy this unity. The geeks of XYZ prepared a special kind of perfume. Anyone who inhales this perfume becomes extremely violent. The students of XYZ somehow manage to spread this perfume throughout ABC's campus atmosphere.

There are *N* boys $(1, 2, 3, \ldots . N)$ and *N* girls $(1, 2, 3, \ldots . N)$ in ABC college. Each boy has a crush on a single girl and each girl has a crush on a single boy. Since the perfume has been inhaled by each and every student of ABC college, every student decides to beat up his/her crush's crush, ie. , if boy *x* has a crush on girl *y* and girl *y* has a crush on boy *z*, *x* will beat *z* up, provided, of course, if *x* and *z* is not the same person.

The doctor of ABC college foresees this situation. He cannot stop so many people from beating each other up, however, he can be prepared for the worst-case patient(s). The worst-case patient(s) will be the patient(s) who get(s) beaten up by the maximum number of students. The doctor comes to you for help. He has *2* questions for you :

1. What is the number of beatings received by the worst-case patient(s) ?

2. What is the total number of pairs of students who ended up beating up each other ?

Constraints :

$1 \leq T \leq 10$

$1 \leq N \leq 10^5$

Input :

The first line comprises of *T*, the number of test cases. Each test case comprises of *3* lines. The first line consists of *N*.

The next line consists of *N* space separated natural numbers between *1* and *N* inclusive such that the ith number denotes the the crush of boy *i*.

The next line consists of *N* space separated natural numbers between *1* and *N* inclusive such that the ith number denotes the the crush of girl *i*.

Output :

For every test case, on a new line, print two space separated integers, the answer to doctor's question *1* followed by answer to doctor's question *2*.

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
2
3
2 2 1
3 2 1
4
2 3 4 1
2 3 4 1
```

EXPECTED OUTPUT

```
1 0
1 4
```

Test Case 2

INPUT (STDIN)

```
2
4
4 2 1 3
4 2 1 2
3
2 4 1
2 4 1
```

EXPECTED OUTPUT

```
1 0
1 1
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
while(true)
```

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

```
4
```

Test Case 2

TOKEN COUNT

```
535
```

Test Case 3

NLOC

```
95
```

Code

C campus_crushes.c

```c
#include <stdio.h>
#include <stdlib.h>

#define MAXN 100005

int boy_crush[MAXN];
int girl_crush[MAXN];
int boy_beats[MAXN];
int girl_beats[MAXN];
int boy_in[MAXN];
int girl_in[MAXN];

int main() {
    int t;
    if (scanf("%d", &t) != 1) return 0;

    while(true) {
        if (t <= 0) break;
        t--;
        int n;
        if (scanf("%d", &n) != 1) break;

        for (int i = 1; i <= n; i++) {
            scanf("%d", &boy_crush[i]);
            boy_in[i] = girl_in[i] = 0;
        }
        for (int i = 1; i <= n; i++) {
            scanf("%d", &girl_crush[i]);
        }

        for (int i = 1; i <= n; i++) {
            int g = boy_crush[i];
            int b = girl_crush[g];
            boy_beats[i] = (i != b) ? b : 0;
            if (boy_beats[i]) boy_in[boy_beats[i]]++;

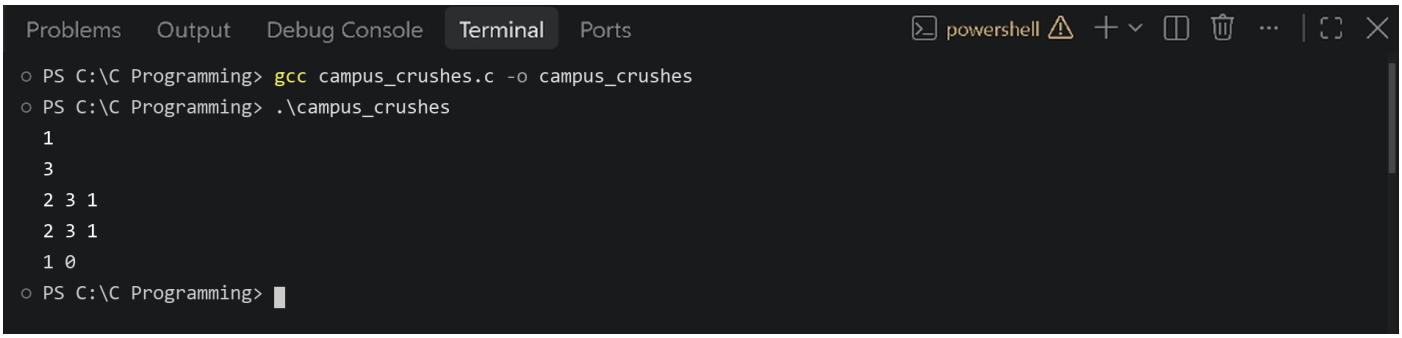
            int target_g = boy_crush[b];
            girl_beats[g] = (g != target_g) ? target_g : 0;
            if (girl_beats[g]) girl_in[girl_beats[g]]++;
        }

        int max_beatings = 0;
        for (int i = 1; i <= n; i++) {
            if (boy_in[i] > max_beatings) max_beatings = boy_in[i];
            if (girl_in[i] > max_beatings) max_beatings = girl_in[i];
        }

        int mutual_pairs = 0;
        for (int i = 1; i <= n; i++) {
            int b = boy_beats[i];
            if (b > i && boy_beats[b] == i) mutual_pairs++;
            int g = girl_beats[i];
            if (g > i && girl_beats[g] == i) mutual_pairs++;
        }

        printf("%d %d\n", max_beatings, mutual_pairs);
    }
    return 0;
}
```

Output


Problems
Output
Debug Console
Terminal
Ports
powershell
PS C:\C Programming> gcc campus_crushes.c -o campus_crushes
PS C:\C Programming> .\campus_crushes
1
3
2 3 1
2 3 1
1 0
PS C:\C Programming>